COMPANY LOGO	Project Title	Some Project	Project No.	26xxx
	Client	Some Client	Calculation No.	26xxx_CAL001
	Calculation Title	Some Calculation	Revision	D
	Project Phase	Detailed Design	Date	27/04/2026

Calculation Objective / Scope

This is a Quarto document template (`.qmd`) intended to be used to develop engineering calculations in the format of a typical engineering calculation pad.

You can learn more about creating custom Typst templates here:

https://quarto.org/docs/output-formats/typst-custom.html

For actual use, the introduction should be a *brief* description of the calculation objective and scope and ideally include a picture.

Key Assumptions

Document key calculation assumptions here. Generally just repeat the assumptions from the assumptions tab, but if there are a lot summarise the most important handful.

- Assumption 1
- Assumption 2
- …

Holds

- Hold 1
- Hold 2
- Hold 3

Key Findings

- Finding 1
- Finding 2
- Finding …

Rev.	Date	Description	Prepared	Checked	Approved
D	27/04/2026	Heaps More preliminary	A. Engineer	B. Engineer	C. Engineer
C	26/04/2026	Even More preliminary	A. Engineer	B. Engineer	C. Engineer
B	25/04/2026	More preliminary	A. Engineer	B. Engineer	C. Engineer

Table of Contents

1 Header 1

1.1 Header 2

1.1.1 Header 3

2 Introduction

2.1 Calculation Objective / Scope

This calculation covers the design of the structural steel and foundation for the XXXX.

The structural steel consists of three individual sub-structures isolated to control vibrations. These buildings are all braced framed systems and columns are considered pinned base.

Foundations are either combined or individual spread reinforced concrete footings.

In addition to the main structural framing, the calculation verifies the structural adequacy of:

- Footings including hold down bolts
- Mechanical frames and equipment plinths
- Associated chute-work and chute supports

2.2 Exclusions

This calculation excludes the following:

- Design of the process / mechanical function for items where structural design has been completed, e.g. chutes, hoppers
- Structural design of vendor supplied equipment

3 Basis of Design

This calculation has been completed in accordance with XXXX. The Basis of Design for this calculation is described in XXXX, with mechanical load data in accordance with the XXXX.

3.1 Design Standards and Codes

Unless specifically noted, the design complies with Australian Standards and selected international standards as listed in the Design Criteria. The following summarises the key standards used in producing this calculation:

- AS 1170.0 - 2002 Structural design actions - Part 0: General principles
- AS 1170.1 - 2002 Structural design actions - Part 1: Permanent, imposed and other actions
- AS 1170.2 - 2021 Structural design actions - Part 2: Wind actions
- AS 1170.4 - 2007 Structural design actions - Part 4: Earthquake Actions in Australia
- AS 3600 - 2018 Concrete structures
- AS 4100 - 1998 Steel structures

3.2 Other Reference Documents

Other reference documents used in this design include:

- Geotechnical Report
- Certified Vendor Drawings

3.3 Client Provided Information

The following information has been provided by the client:

- Something
- Something else.

3.4 Holds

4 Work Records

4.1 Design Calculations

The following design calculations have been relied upon in the preparation of this calculation:

Doc Number	Title	Revision	Link / Path
26xxx_CAL002	Structural Steel Design	A	link
26xxx_CAL003	Foundation Design	A	link

4.2 Output

This calculation covers the structural components on the drawings and/or sketches listed in the following table:

Drawing Number	Drawing Title	Revision
0001	General Arrangement	A
0002	Details 1	0
0003	Details 2	C1
0004	Earthworks	x

5 Loads

This section describes the loads applied.

5.1 Gravity Loads

Document gravity loads here.

5.1.1 GSW Structural Self Weight

Structural self weight is applied as an acceleration in the model unless noted otherwise.

5.1.2 GFM Flooring

Floor mesh self weight is assumed to be:

$$mass = 50\,\mathrm{kg/m^2}$$

$$GFM = mass \times g_{acc} = 0.5\,\mathrm{kPa}$$

Flooring is applied as follows:

- A dummy load case of 1kPa is applied to all floor areas.
- The actual floor load is then applied as a load combination of 0.5 x the dummy case.

5.1.3 GHR Handrails

Handrails are assumed to be 15kg/m.

$$GHR = 15\ \mathrm{kg/m} \times g_{acc} = 1.5\ \mathrm{kN/m}$$

5.1.4 GME Mechanical Equipment

Break into multiple load cases if required.

5.2 Live Loads

5.2.1 QLL5 Standard Floor Live Load

5.2.2 QME Mechanical Equipment Live Load

Break into multiple load cases if required.

5.3 Load Combinations

6 Analysis Model

The following figure shows the analysis model:

6.1 Restraints, Releases & Offsets

The following restraints, releases and offsets have been used:

- All columns assumed to be pinned at base unless otherwise noted.
- All members assumed to be pinned at their ends unless otherwise noted.
- Members modelled at their centrelines unless otherwise noted.

6.2 Analysis Method

The following analysis methods have been used:

- Non-linear static analysis.
- Linear buckling where required to determine effective lengths for members.

6.3 Steel / Concrete Design Assumptions

The following steel and concrete design assumptions have been used:

- …

7 Summary

We have done some awesome calcs.

8 Calculation

Copy this section as required.

8.1 Example Calcs

Do some calculation here.

```
from cycler import cycler
import numpy as np
import polars as pl
import matplotlib.pyplot as plt

a = 1
b = 2
a + b
```

```
3
```

And now plot a graph (see Figure 1):

```
no_points = 101
jitter_scale = 0.2
x = np.linspace(0, 2 * np.pi, no_points)
y1 = np.sin(x) + (np.random.random(no_points) - 0.5) * jitter_scale
y2 = np.cos(x) + (np.random.random(no_points) - 0.5) * jitter_scale
y3 = np.sin(x * 2) + (np.random.random(no_points) - 0.5) * jitter_scale
y4 = np.cos(x * 2) + (np.random.random(no_points) - 0.5) * jitter_scale
y5 = np.sin(x * 3) + (np.random.random(no_points) - 0.5) * jitter_scale
y6 = np.cos(x * 3) + (np.random.random(no_points) - 0.5) * jitter_scale
y7 = np.sin(x * 4) + (np.random.random(no_points) - 0.5) * jitter_scale
y8 = np.cos(x * 4) + (np.random.random(no_points) - 0.5) * jitter_scale
```

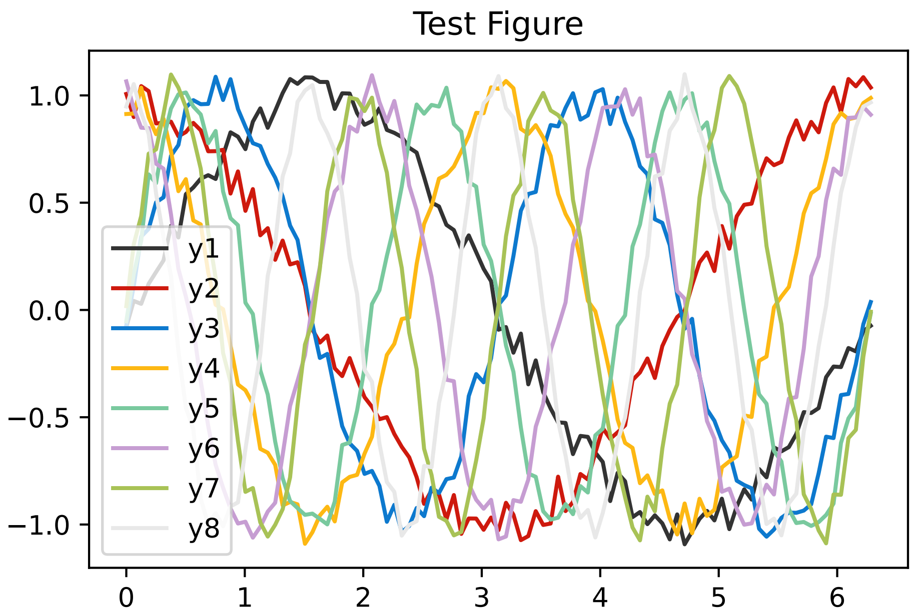

Figure 1: Test Figure

Woohoo!

And do some data processing with Polars (see Table 1):

```
df = pl.DataFrame({"x": x, "y": y1, "z": y2})

pl.Config.set_tbl_hide_column_data_types(True)
pl.Config.set_float_precision(3)

df.head()
```

Table 1: Test Data

x	y	z
0.000	−0.079	1.007
0.063	0.042	0.899
0.126	0.028	1.043
0.188	0.122	1.020
0.251	0.177	0.869

Tip 1: Tip

This is an example of a callout “tip”.

Warning

This is an example of a callout “warning”.

! Important

This is an example of an “important” callout.

Callouts can even be cross-referenced - see Tip 1. So can sections - see Section 2.

9 Appendix A

10 Appendix B